

Literature Resolution: The Limits in the Mean Counting Function Cannot Be Interchanged

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13 August 2026

Status: literature already answered (fully, negatively). Problem 1 of arXiv:2005.05094 asks whether one may interchange the $T \rightarrow \infty$ and $\sigma \rightarrow 0^+$ limits in the mean counting function for every Dirichlet series in the relevant Nevanlinna class. Theorems 1.6 and 1.7 of arXiv:2405.03522 give two counterexamples: equality can fail even when the interchanged limit exists, and the interchanged limit itself can fail to exist.

1 The source question

O. F. Brevig and K.-M. Perfekt, *A mean counting function for Dirichlet series and compact composition operators*, arXiv:2005.05094, define for a Dirichlet series f the mean counting function by first restricting the zeros to $\operatorname{Re} s > \sigma$, averaging in the vertical direction, and then taking $\sigma \rightarrow 0^+$. Section 8.1 asks whether the order of these limits can always be reversed. In the notation of the paper:

Problem 1 in the source paper. *Let $f \in \mathcal{N}_u$. Is it true that*

$$M_f(w) = \lim_{T \rightarrow \infty} \frac{\pi}{T} \sum_{\substack{s \in f^{-1}(\{w\}) \\ |\operatorname{Im} s| < T \\ 0 < \operatorname{Re} s < \infty}} \operatorname{Re} s$$

for every $w \neq f(+\infty)$?

Writing the expression before the T -limit as $N_f(w, T)$, the question is whether

$$\lim_{T \rightarrow \infty} N_f(w, T) = M_f(w)$$

always holds.

2 The later resolution

O. F. Brevig, A. Kouroupis, and K.-M. Perfekt, *Almost periodicity and boundary values of Dirichlet series*, arXiv:2405.03522, explicitly identifies the question above as “[12, Problem 1]” and proves:

Later counterexamples (Theorems 1.6 and 1.7). *For every $0 < \varepsilon < 1$ there is a Dirichlet series $f : \mathbb{C}_0 \rightarrow \mathbb{D}$, with $f(+\infty) = 0$, such that*

$$\lim_{T \rightarrow \infty} N_f(\xi, T) = \log \frac{1}{|\xi|}$$

for quasi-every $\xi \in \mathbb{D} \setminus \{0\}$, whereas

$$M_f(\xi) \leq \log \frac{1}{|\xi|} - (1 - \varepsilon) \frac{1 - |\xi|^2}{2} \quad (\xi \neq 0).$$

Thus the proposed equality fails wherever the first display holds.

Moreover, for every $0 < \varepsilon < 1$ there is a Dirichlet series $f : \mathbb{C}_0 \rightarrow \mathbb{D}$ for which $\lim_{T \rightarrow \infty} N_f(\xi, T)$ fails to exist for quasi-every ξ satisfying $\varepsilon < |\xi| < 1$.

These are Theorems 1.6 and 1.7 of the later paper. Immediately after their statements, the authors write that those two theorems resolve the earlier Problem 1.

3 Why this answers the exact question

The later examples are bounded Dirichlet series mapping the right half-plane to the unit disc, hence lie within the source paper's class $\mathcal{N}_u$. The quantity $N_f(\xi, T)$ is exactly the uncut finite-height sum appearing in the source problem. Theorem 1.6 refutes the universal equality while retaining existence of the T -limit; Theorem 1.7 gives the stronger failure of existence. Therefore the answer to Problem 1 is unambiguously *no*.

No new mathematical result is claimed in this literature packet. The same target also produced a separate new partial-result packet for the source paper's weighted boundary-embedding problem.

References

- [1] O. F. Brevig and K.-M. Perfekt, *A mean counting function for Dirichlet series and compact composition operators*, Mathematische Annalen 381 (2021), 1367–1402; arXiv:2005.05094.
- [2] O. F. Brevig, A. Kouroupis, and K.-M. Perfekt, *Almost periodicity and boundary values of Dirichlet series*, Transactions of the American Mathematical Society 378 (2025), 5677–5700; arXiv:2405.03522; doi:10.1090/tran/9420.